MIL-T-12089E
19 November 1981
SUPERSEDING
MIL-T-12089D
13 January 1976

MILITARY SPECIFICATION

TOWER, DOUBLE-STRUCTURE, SURVEYING: COMPLETE, WITH ANCHORS, BOLTS, AND ACCESSORIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a double-structure surveying tower.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5445

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PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000 Pound Capacity.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Water Proofing; with Appropriate Test Methods.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13215E1737 - Tower, Double Structure, Surveying.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ALUMINUM ASSOCIATION (AA)

Specifications for Aluminum Structures Construction Manual Series

(Application for copies should be addressed to the Aluminum Association, 818 Connecticut Avenue, NW, Washington, DC 20006.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, Incorporated, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Technical society of technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The surveying tower shall be in accordance with Top Assembly TA13215E1737, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit and assembly of the tower. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 6.3).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more towers for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings (see 6.8).

3.5 Threads. All screw threads shall conform to FED-STD-H28.

3.6 Safety. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided.

3.7 Anchor post assemblies and steps. An unassembled set of six anchor post assemblies and 13 steps shall be furnished with each unit in addition to the quantity shown on the drawings.

3.8 Bolts, nuts, and washers. Additional bolts, nuts, and washers equal in quantity to those called for in the parts list shall be furnished with each tower.

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3.9 Erection markings.

3.9.1 Die stamping. Each structural member of the tower shall be stamped with steel dies using Gothic-type capitals and Arabic numerals as shown on the drawings. Letters and numerals shall be not less than 1 inch high.

3.9.2 Stenciling. As shown on the drawings, the stenciling of the platform deckings shall be in accordance with TT-I-1795, type I, color black.

3.10 Chest. The chest shall be as shown on the Top Assembly TA13215E1737 and as specified herein.

3.11 Identification marking. Each tower shall be identified in accordance with MIL-STD-130. The contractor's identification shall be placed on not less than two structural members near the top and bottom of the tower.

3.12 Workmanship.

3.12.1 Aluminum fabrication. Aluminum used in the fabrication of the tower shall be free from kinks and sharp bends. The straightening and bending of the material shall be done by methods that will not cause damage to the metal. All bends shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum shall be cut by shearing, routing, or by highspeed saws. Shearing and sawing shall be done neatly and accurately. Unless otherwise indicated, corners shall be square and true. All burrs and rough edges shall be ground smooth. The surface and edges of all finished members shall be smooth and free from kinks, bends, winds, and excessive scratches.

3.12.2 Welding. Welding shall be in accordance with AA Specification for Aluminum Structures Construction Manual Series, Section 1, Welding Fabrication.

3.12.3 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be corrrectly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

3.12.4 Assembly. The tower components shall be fabricated to assure ease of assembly, fit, and interchangeability of like components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or another facilities suitable for the the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly Inspection. Failure of any examination or test by the first article shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawing in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Initial production inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Inspection comparison (see 4.7).

e. Inspection of packaging (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article tower shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first article tower shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production towers will be selected at random by the Government from the towers being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production tower shall not exclude the remaining towers from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

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4.4.1 Inspection failure. Failure of an initial production tower to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production tower and shall be cause for refusal by the Government to continue acceptance of production towers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on towers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all towers accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling of tower components for examination and test shall be in accordance with MIL-STD-105, inspection level S-2.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the defects specified in 4.6.1. AQL shall be 2.5 percent defective.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The tower shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Dimensions and tolerance not as shown on the drawings.
103. Member not true to dimension and line.
104. Safety devices not as specified.
105. Additional anchor post assemblies and steps not furnished.
106. Additional bolts, nuts, and washers not furnished.
107. Identification and other required markings not as specified.
108. Workmanship not as specified.

4.6.2 Tests.

4.6.2.1 Assembly. The tower shall be erected with all parts and members fitted to their correct positions. During erection, the tower shall be inspected for ease of assembly and fit. Evidence of any member which cannot be placed without forcing shall constitute failure of this test.

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4.6.2.2 Instrument fasteners. The instrument fasteners shall be tested for conformance to FED-STD-H28 with certified thread ring gages. Evidence of any fastener not mating with the gage shall constitute failure of this test.

4.7 Inspection comparison. The Government may select towers at any time during the contract production period and subject these towers to the examination specified in 4.6.1 and the tests specified in 4.6.2, to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on towers selected at random from those which have been accepted by the Government and will not include the previously inspected first article and initial production towers. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison tower to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison tower and shall be cause for refusal by the Government to continue acceptance of production towers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on towers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all towers accepted prior to the completion of inspection comparison are similarily deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction (first article) pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined in accordance with 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-2.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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109. Materials, methods, and containers not as specified. Each incorrect material, method, or container shall constitute one defect.
110. Preservatives not applied as specified for level A.
111. Fiberboard boxes not closed as specified for level A.
112. Bundling and boxing of items not in accordance with tables I and II.
113. Strapping not as specified for level A.
114. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preproduction (first article) pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be that specified in section 4 and shall be subjected to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the preproduction model or a production tower. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production towers.

5.2 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Unprotected surfaces. Unpainted exterior metal surfaces of bolts, nuts, washers, hooks, U-bolts, steps, and any other exterior metal surface not listed herein, requiring the application of a contact preservative in accordance with MIL-P-116 shall be cleaned, dried, and coated with type P-1 preservative conforming to the applicable specification listed in MIL-P-116.

5.2.1.2 Belts. The belts shall be placed in a fiberboard box conforming to PPP-B-636, W6c and the box closed in accordance with method V of the box specification appendix.

5.2.1.3 Small loose components. Small loose components of like size and description such as bolts, nuts, washers, hooks, U-bolts, and plates shall be placed in close-fitting boxes conforming to PPP-B-636, W5c. Closure of the box shall be as specified in 5.2.1.2. Small loose components shall, to the extent possible, be consolidated in the chest. Preserved parts which will not fit in the chest shall be consolidated in boxes conforming to PPP-B-601, domestic type, style optional, grade B, in quantities not to exceed the weight limitation of the container. Blocking, bracing, and cushioning shall be provided to prevent movement and damage to the components and the containers.

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5.2.1.4 Other loose components. Other loose components, except those specified in 5.2.1.2, shall be placed in close-fitting boxes conforming to PPP-B-601, domestic type, style optional, grade B, in quantities not to exceed the weight limitations of the container. Blocking, bracing, and cushioning shall be provided to prevent movement and damage to the components and the containers.

5.2.1.5 Structural components. Structural components shall be bundled in uniform compact bundles in accordance with table I and II and the bundles shall be strapped with 1-inch by 0.038-inch strapping conforming to QQ-S-781, class 1, type I or IV and unless otherwise specified, shall be finish B (see 6.2). When specified (see 6.2), finish A strapping shall be used. The two end straps shall be placed approximately one-sixth the distance from each end of the bundle. Intermediate straps as required by tables I and II shall be placed equidistant between the two end straps. Other components shall be boxed or loose as indicated in tables I or II. Boxes shall conform to PPP-B-601, domestic type, style optional.

5.2.1.6 Erection drawings, manuals, and technical publications. The erection drawings, manuals, and technical publications for each tower shall be preserved in accordance with MIL-P-116, method IC-1. The preserved drawings, manuals, and publications for the applicable tower shall be secured within the chest in a protected location.

5.2.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.3 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.3.1 Level A. The structural components, boxed and loose components, including the chest, comprising one complete tower, shall be packed in a crate conforming to MIL-C-3774, type I, style A. Arrangement shall provide maximum compactness, minimum voids, and good balance. Blocking, bracing, and anchoring shall be in accordance with MIL-STD-1186. Strapping shall conform to QQ-S-781, class 1, type I or IV size as applicable, and, unless otherwise specified, shall be finish B (see 6.2). When specified (see 6.2), finish A strapping shall be used.

5.3.2 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.4 Marking. Bundles, boxes, loose components and shipping containers shall be in accordance with MIL-STD-129 for level A protection. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tower is intended for use in large scale mapping operations. The inner tower supports the surveying instrument. The outer tower supports a platform for the observer and his notekeeper, and a higher platform for the light tender and his lamps.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and number of units required (see 3.2).

c. When initial production inspection is required and number of towers to be furnished (see 3.3).

d. Time frame required for submission of preproduction pack (see 5.1).

e. Degree of preservation and degree of packing required (see 5.2 and 5.3).

f. When other than finish B strapping is required (see 5.2.1.5 and 5.3.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a preproduction model. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Preproduction (first article) pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, package, pack, and mark the towers in accordance with this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each tower.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of towers.

6.7 Army Technical Manual TM 5-9413. Department of the Army Technical Manual TM 5-9413, Tower, Triangulation, Structural Aluminum, 103 Foot, covers the erection, dismantling, and re-erection of this tower.

6.8 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Custodians:
Army - ME
Air Force - 99

Preparing activity:
Army - ME

Project 5445-0016

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TABLE I. Packaging of members for outer tripod

Type	Mark No.	Description and size	Length Ft	Length In	No. of pieces	Number of straps
		Chest of bolts and tools			1	2
Bundle	0 1 TF	Anchor post	4	11-1/2	12	3
	OT-1	Leg 3-1/2x3-1/2x1/4-60° L	13	9-1/4	3	
	OT-1-H*	Horizontal 3x3x1/4 L	18	2-1/4	3	
	OT-1-H*	Horizontal 2-1/2x2-1/2x 3/16 L	19	5-7/16	3	
Bundle	OT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	11	6-9/16	3	4
	OT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	12	5/16	3	
	OT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	10	9-7/16	3	
	OT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	12	9-7/16	3	
	OT-1-V	Vertical 1-1/2x1-1/2x3/16 L	6	3	3	
Box		Steps 3/4 Ø Rod	1	0	12	
	OT-2	Leg 3-1/2x3-1/2x1/4-60° L	13	9-1/4	3	
	OT-2-H*	Horizontal 3x3x3/16 L	15	8-11/16	3	
	OT-2-H*	Horizontal 2-1/2x2-1/2x 3/16 L	15 16	11-7/8	3	
Bundle	OT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	10	5	3	
	OT-2-D	Diagonal 1-1/2x1-1/2/3/16 L	11	1-13/16	3	4
	OT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	9	8-1/2	3	
	OT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	11	10-5/16	3	
	OT-2-V	Vertical 1-1/2x1-1/2x3/16 L	6	2-1/4	3	
Box		Steps 3/4 Ø Rod	1	0	13	
	OT-3	Leg 3-1/2x3-1/2x1/4-60° L	13	9-1/4	3	
	OT-3-H	Horizontal 2-1/2x2-1/2x 3/16 L	14	1/8	3	
	OT-3-H*	Horizontal 2x2x3/16 L	14	10-3/4	3	
	OT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	9	8-15/16	3	
Bundle	OT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	10	2-13/16	3	4
	OT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	9	1-7/16	3	
	OT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	10	10-5/16	3	
	OT-3-V	Vertical 1-1/2x1-1/2x3/16 L	6	3-5/8	3	
Box		Steps 3/4 Ø Rod	1	0	13	

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TABLE I. Packaging of members for outer tripod (Continued)

Type	Mark No.	Description and size	Length Ft	Length In	No. of pieces	Number of straps
	OT-4	Leg 3x3x1/4-60° L	13	9-1/4	3	
	OT-4-H	Horizontal 2-1/2x2-1/2x 3/16 L	12	3-3/4	3	
Bundle	OT-4-H	Horizontal 2x2x3/16 L	13	2-3/8	3	
	OT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	9	7-5/16	3	
	OT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	9	1-5/8	3	4
	OT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	10	2-5/16	3	
	OT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	8	6-5/8	3	
Box		Steps 3/4 Ø Rod.....	1	0	13	
	OT-5	Leg 3x3x3/16-60° L	13	9-1/4	3	
	OT-5-H	Horizontal 2-1/2x2-1/2x 3/16 L	11	6	3	
Bundle	OT-5-H	Horizontal 2-1/2x2-1/2x 3/16 L	10	7-3/8	3	
	OT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	13	6-1/2	3	
	OT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	13	6-1/2	3	
	OT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	12	9-7/8	3	4
	OT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	12	9-7/8	3	
	OT-5-V	Vertical 1-1/2x1-1/2x3/16 L	3	9-3/16	3	
	OT-5-V	Vertical 1-1/2x1-1/2x3/16 L	3	8-1/4	3	
Box		Steps 3/4 Ø Rod	1	0	13	
	OT-6	Leg 2-1/2x2-1/2x3/16-60° L	13	9-1/4	3	
	OT-6-H	Horizontal 2-1/2x2-1/2x 3/16 L	9	9-3/8	3	
	OT-6-H	Horizontal 2x2x3/16 L	8	10-3/4	3	
	OT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	12	11/16	3	
Bundle	OT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	12	11/16	3	4
	OT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	11	4-5/8	3	
	OT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	11	4-5/8	3	
	OT-7	Leg 2-1/2x2-1/2x3/16-60° L	13	9-1/4	3	
	OT-7-H	Horizontal 2x2x3/16 L	8	3/4	3	
Bundle	OT-7-H	Horizontal 1-1/2x1-1/2x 3/16 L	7	2-1/8	3	
	OT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	10	7-3/4	3	4
	OT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	10	7-3/4	3	
	OT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	10	3/8	3	
	OT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	10	3/8	3	

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TABLE I. Packaging of members for outer tripod (Continued)

Type	Mark No.	Description and size	Length Ft	Length In	No. of pieces	Number of straps
Bundle	OT-8	Observer's platform			3	3
	OT-8	Leg 2-1/2x2-1/2x3/16-60° L	11	1-1/2	3	
	OT-8-H	Horizontal 1-1/2x1-1/2 3/16 L	3	10-7/8	3	
	OT-8-H	Floor Brace 1-1/2x1-1/2x 3/16 L (Bent)	6	10-1/4	3	
	OT-8-H	Railing 2x2x3/8 L	6	7-3/16	6	
	OT-8-H	Railing 1-1/2x1-1/2x 3/16 L (Bent)	6	7-3/16	3	3
Bundle	OT-8-D	Diagonal 1-1/2x1-1/2x3/16 L	9	2-3/16	3	
	OT-8-D	Diagonal 1-1/2x1-1/2x3/16 L	9	2-3/16	3	
	OT-8-D	Tie 1-1/2x1-1/2x3/16 L (Bent left)	5	5-5/8	3	
	OT-8-D	Tie 1-1/2x1-1/2x3/16 L .... (Bent right)	5	5-5/8	3	
	OT-8-V	Tie 1-1/2x1-1/2x3/16 L	4	7-1/2	3	
Loose	OT-9	Light keeper's platform			1	
	OT-9	Leg 2-1/2x2-1/2x3/16-60° L	7	7-1/8	3	
	OT-9-H	Lintel 1-1/2x1-1/2x3/16 L (Bent)	6	3-13/16	3	
	OT-9-H	Horizontal 1-1/2x1-1/2x 3/16 L	3	10-7/8	3	
Bundle	OT-9-D	Diagonal 1-1/2x1-1/2x 3/16 L (Bent left)	3	11-3/8	3	
	OT-9-D	Diagonal 1-1/2x1-1/2x3/16 L (Bent right)	3	11-3/8	3	3
	OT-9	Swivel leg holder			6	
	OT-9	Swivel leg			3	
	OT-9	Peak (welded assembly)			1	
	OT-9	Top plate 15' Øx1/4			1	
	OT-9	Top plate angles 1-1/2x 1-1/2x3/16 L		12	3	
Box		Steps 3/4 Ø Rod	1	0	40	
Box		Steps 3/4 Ø Rod	1	0	13	

*Members marked with asterisk (*) are made up in field with splice connection. Each part shipped "knocked-down" as two pieces and splicing angle.

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TABLE II. Packaging of members for inner tripod

Type	Mark No.	Description and size	Length Ft	Length In	No. of pieces	Number of straps
	IT-1	Leg 3-1/2x3-1/2x1/4-60° L	13	9-1/4	3	
	IT-1-H*	Horizontal 2-1/2x2-1/2x 3/16 L	16	1-7/8	3	
	IT-1-H*	Horizontal 2-1/2x2-1/2x 3/16 L	17	5-1/8	3	
Bundle	IT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	10	8-3/16	3	4
	IT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	11	2-7/8	3	
	IT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	9	8-3/16	3	
	IT-1-D	Diagonal 1-1/2x1-1/2x3/16 L	12	2-7/8	3	
	IT-1-V	Vertical 1-1/2x1-1/2x3/16 L	6	2-7/16	3	
	IT-2	Leg 3-1/2x3-1/2x1/4-60° L	13	9-1/4	3	
	IT-2-H	Horizontal 2-1/2x2-1/2x 3/16 L	13	8-5/16	3	
Bundle	IT-2-H*	Horizontal 2x2x3/16 L	14	11-9/16	3	4
	IT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	9	8-1/4	3	
	IT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	10	3-7/8	3	
	IT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	8	8-1/16	3	
	IT-2-D	Diagonal 1-1/2x1-1/2x3/16 L	11	4-1/16	3	
	IT-2-V	Vertical 1-1/2x1-1/2x3/16 L	6	1-1/2	3	
	IT-3	Leg 3x3x1/4-60° L	13	9-1/4	3	
	IT-3-H	Horizontal 2-1/2x2-1/2x 3/16 L	11	2-3/4	3	
Bundle	IT-3-H	Horizontal 2x2x3/16 L	12	5-15/16	3	
	IT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	8	8-9/16	3	4
	IT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	9	6-1/4	3	
	IT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	7	9-9/16	3	
	IT-3-D	Diagonal 1-1/2x1-1/2x3/16 L	10	5-1/4	3	
	IT-4	Leg 3x3x1/4-60° L	13	9-1/4	3	
	IT-4-H	Horizontal 2x2x3/16 L	8	9-3/16	3	
	IT-4-H	Horizontal 1-1/2x1-1/2x 3/16 L	10	3/8	3	4
	IT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	7	9-9/16	3	
	IT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	8	10-1/8	3	
	IT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	6	11-1/2	3	
	IT-4-D	Diagonal 1-1/2x1-1/2x3/16 L	9	8-3/16	3	

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TABLE II. Packaging of members for inner tripod (continued)

Type	Mark No.	Description and size	Length Ft	Length In	No. of pieces	Number of straps
	IT-5	Leg 3x3x3/16-60° L	13	9-1/4	3	
	IT-5-H	Horizontal 1-1/2x1-1/2x 3/16 L	6	3-5/8	3	
Bundle	IT-5-H	Horizontal 1-1/2x1-1/2x 3/16 L	7	3-13/16	3	4
	IT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	9	6-1/8	3	
	IT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	9	6-1/8	3	
	IT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	10	4-11/16	3	
	IT-5-D	Diagonal 1-1/2x1-1/2x3/16 L	10	4-11/16	3	
	IT-6	Leg 3x3x3/16-60° L	13	9-1/4	3	
	IT-6-H	Horizontal 1-1/2x1-1/2x 3/16 L	3	9-13/16	3	
Bundle	IT-6-H	Horizontal 1-1/2x1-1/2x 3/16	5	1	3	3
	IT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	7	10-3/8	3	
	IT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	7	10-3/8	3	
	IT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	8	7-1/16	3	
	IT-6-D	Diagonal 1-1/2x1-1/2x3/16 L	8	7-1/16	3	
	IT-7	Leg 2-1/2x2-1/2x3/16-60° L	13	8-5/8	3	
	IT-7-H	Horizontal 1-1/2x1-1/2x 3/16	1	4	3	
Bundle	IT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	2	4-5/16	3	3
	IT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	2	6-11/16	3	
	IT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	3	2-3/16	3	
	IT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	3	5-13/16	3	
	IT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	3	11-13/16	3	
	IT-7-D	Diagonal 1-1/2x1-1/2x3/16 L	4	5-13/16	3	
Loose	IT-8	Top section (Welded assembly)			1	

*Members marked with asterisk (*) are made up in field with splice connection. Each part shipped "knocked-down" as two pieces and slicing angle.

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